SQL FOR DATA ANALYSIS · LESSON 11 OF 15

OFFSET-FETCH and Window Functions I: OVER, PARTITION BY & Ranking

Page through results with OFFSET and FETCH, then learn window functions, the most powerful analytics feature in SQL. Understand OVER and PARTITION BY, rank rows with ROW_NUMBER, RANK, DENSE_RANK and NTILE, solve top-N-per-group and de-duplication problems, and compare aggregate windows with GROUP BY.

SQL class 5 · SQL Server 2022, 2026 · Intermediate–Advanced · 130 min

By the end of this lesson you can

- Paginate with ORDER BY ... OFFSET ... ROWS FETCH NEXT ... ROWS ONLY
- Explain what a window function is and how OVER, PARTITION BY and ORDER BY define the window
- Use ROW_NUMBER, RANK, DENSE_RANK and NTILE and explain how ties behave
- Solve top-N per group, Nth highest value and duplicate-removal problems
- Use aggregate window functions to show totals and shares beside detail rows

Datasets used

`superstore_sales.csv` `hr_employees.csv`

OFFSET and FETCH: pagination

`OFFSET n ROWS FETCH NEXT m ROWS ONLY` skips n rows and returns the next m. It requires `ORDER BY`. It is the standard (ANSI) way to page results, used for "page 3 of results" in applications and for "rows 11 to 20" questions.

SQL

```sql
SELECT customer_name, city, segment
FROM customers
ORDER BY customer_id
OFFSET 10 ROWS FETCH NEXT 5 ROWS ONLY;
```

RESULT

customer_name	city	segment
Sai Iyer	Indore	Consumer
Neha Kulkarni	Bengaluru	Consumer
Diya Khan	Bhubaneswar	Home Office
Kabir Das	Kolkata	Consumer
Reyansh Joshi	Mumbai	Corporate

5 rows × 3 columns

Page	OFFSET	FETCH
1	0	page size
2	page size	page size
page p	(p − 1) × page size	page size

`TOP n` cannot skip rows; OFFSET-FETCH can. PostgreSQL/MySQL write `LIMIT m OFFSET n`. On very deep pages (OFFSET 1,000,000) databases still read and discard all skipped rows, so large systems use **keyset pagination** (`WHERE id > last_seen_id ORDER BY id`).

The "Nth highest value" question can be answered with it too: the second-highest line value is `ORDER BY sales DESC OFFSET 1 ROWS FETCH NEXT 1 ROWS ONLY` on distinct values:

SQL

```sql
SELECT DISTINCT sales AS second_highest_sales
FROM superstore
ORDER BY sales DESC
OFFSET 1 ROWS FETCH NEXT 1 ROWS ONLY;
```

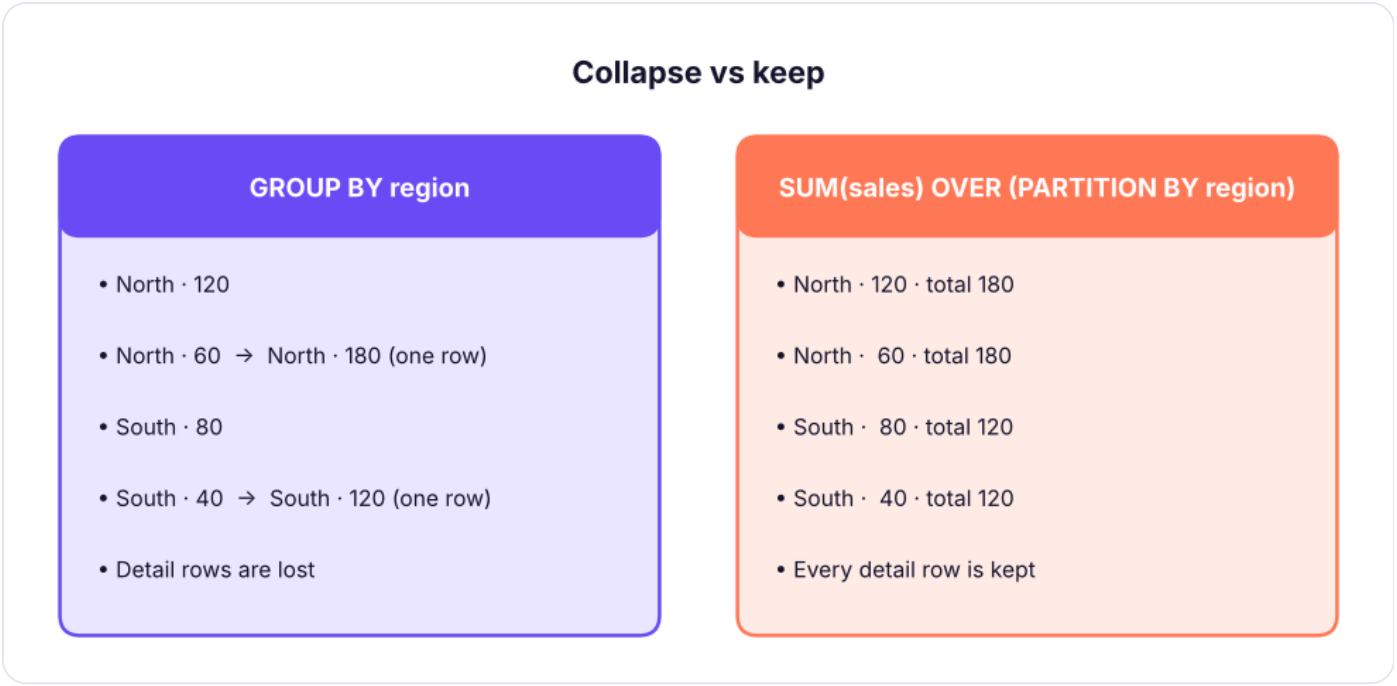
RESULT

second_highest_sales
551,530

1 rows × 1 columns

What is a window function?

A **window function** computes a value for **each row** using a set of related rows (its **window**), **without collapsing the rows** like `GROUP BY` does.



GROUP BY collapses rows into one per group. A window function keeps every row and adds the group's result beside it.

The OVER clause

```sql
function_name(expression) OVER (
    PARTITION BY column_list      -- split rows into groups (optional)
    ORDER BY column_list          -- order inside each group (needed for ranking, running totals)
    ROWS BETWEEN … AND …          -- optional frame (Lesson 12)
)
```

- **`PARTITION BY`** is like GROUP BY for the window: the function restarts for every partition. Without it, the whole result is one partition.
- **`ORDER BY`** inside OVER orders rows *within the partition* (it does not sort the final output; use the normal ORDER BY for that).

```sql
SELECT region, city, order_id, sales,
       SUM(sales) OVER (PARTITION BY region)                        AS region_total,
       ROUND(100.0 * sales / SUM(sales) OVER (PARTITION BY region), 4) AS pct_of_region,
       AVG(sales) OVER ()                                          AS overall_avg
FROM superstore
ORDER BY region, sales DESC;
```

RESULT

region	city	order_id	sales	region_total	pct_of_region	overall_avg
East	Kolkata	IN-2025-03160	472,740	46,484,119	1.02	31,854
East	Kolkata	IN-2025-02134	442,350	46,484,119	0.95	31,854
East	Kolkata	IN-2025-02822	422,640	46,484,119	0.91	31,854
East	Guwahati	IN-2025-02562	413,040	46,484,119	0.89	31,854
East	Patna	IN-2023-00171	393,950	46,484,119	0.85	31,854
East	Patna	IN-2025-02317	388,320	46,484,119	0.84	31,854
East	Bhubaneswar	IN-2025-02981	372,720	46,484,119	0.80	31,854
East	Patna	IN-2024-01792	372,720	46,484,119	0.80	31,854

6,323 rows × 7 columns (first 8 shown)

Each row keeps its own detail but now also shows its region total, its share of the region, and the overall average. With GROUP BY you would need a join to get this.

Ranking functions

All four need `ORDER BY` inside `OVER`.

Function	Behaviour with ties	Example values 90, 80, 80, 70
`ROW_NUMBER()`	Unique numbers; ties broken arbitrarily	1, 2, 3, 4
`RANK()`	Ties share a rank; **gaps** follow	1, 2, 2, 4
`DENSE_RANK()`	Ties share a rank; **no gaps**	1, 2, 2, 3
`NTILE(n)`	Splits rows into n equal buckets	4 rows, NTILE(2): 1, 1, 2, 2

SQL

```sql
WITH scores AS (
    SELECT 'Asha' AS name, 90 AS score UNION ALL SELECT 'Ravi', 80 UNION ALL SELECT
'Meera', 80 UNION ALL SELECT 'Kabir', 70
)
SELECT name, score,
       ROW_NUMBER() OVER (ORDER BY score DESC) AS row_number,
       RANK()       OVER (ORDER BY score DESC) AS rank,
       DENSE_RANK() OVER (ORDER BY score DESC) AS dense_rank,
       NTILE(2)     OVER (ORDER BY score DESC) AS half
FROM scores;
```

RESULT

name	score	row_number	rank	dense_rank	half
Asha	90	1	1	1	1
Meera	80	2	2	2	1
Ravi	80	3	2	2	2
Kabir	70	4	4	3	2

4 rows × 6 columns

Ravi and Meera tie at 80: `RANK` gives both 2 and skips 3; `DENSE_RANK` gives both 2 and continues with 3; `ROW_NUMBER` gives 2 and 3 in arbitrary order (add a tie-breaker column to make it deterministic).

Ranking within groups

"Rank each sub-category by sales within its category":

SQL

```sql
SELECT category, sub_category,
       ROUND(SUM(sales), 0)                                          AS sales,
       RANK() OVER (PARTITION BY category ORDER BY SUM(sales) DESC)  AS
rank_in_category
FROM superstore
GROUP BY category, sub_category
ORDER BY category, rank_in_category;
```

RESULT

category	sub_category	sales	rank_in_category
Furniture	Tables	14,046,224	1
Furniture	Bookcases	11,274,849	2
Furniture	Chairs	8,091,474	3
Furniture	Furnishings	1,429,307	4
Office Supplies	Appliances	6,333,529	1
Office Supplies	Storage	2,659,340	2
Office Supplies	Paper	493,232	3
Office Supplies	Binders	401,219	4
Office Supplies	Art	245,985	5
Technology	Copiers	78,749,926	1
Technology	Machines	42,590,335	2
Technology	Phones	31,919,223	3

13 rows × 4 columns (first 12 shown)

Note the order: `GROUP BY` runs first, then the window function ranks the grouped rows. That is why `SUM(sales)` can appear inside `OVER`.

Top-N per group

You cannot filter on a window function in `WHERE` (windows are computed in the SELECT step, after WHERE). Wrap the query in a CTE or derived table, then filter the rank.

"Top 2 cities by sales in each region":

SQL

```sql
WITH city_sales AS (
    SELECT region, city, SUM(sales) AS sales
    FROM superstore
    GROUP BY region, city
), ranked AS (
    SELECT region, city, sales,
           DENSE_RANK() OVER (PARTITION BY region ORDER BY sales DESC) AS rnk
    FROM city_sales
)
SELECT region, city, ROUND(sales, 0) AS sales, rnk
FROM ranked
WHERE rnk ⩽ 2
ORDER BY region, rnk;
```

RESULT

region	city	sales	rnk
East	Kolkata	13,888,466	1
East	Patna	12,444,393	2
North	Delhi	15,657,610	1
North	Chandigarh	11,541,792	2
South	Kochi	14,713,126	1
South	Bengaluru	13,625,216	2
West	Pune	17,357,038	1
West	Indore	14,408,485	2

8 rows × 4 columns

"Each customer's most recent order" (use ROW_NUMBER to get exactly one row per customer even with ties):

SQL

```sql
WITH ranked AS (
    SELECT customer_id, order_id, order_date,
           ROW_NUMBER() OVER (PARTITION BY customer_id ORDER BY order_date DESC, order_id
DESC) AS rn
    FROM orders
)
SELECT customer_id, order_id, order_date
FROM ranked
WHERE rn = 1
ORDER BY customer_id;
```

RESULT

customer_id	order_id	order_date
C0001	IN-2025-02263	2025-04-09
C0002	IN-2025-02871	2025-10-21
C0003	IN-2025-02642	2025-08-18
C0004	IN-2025-02344	2025-05-10
C0005	IN-2025-02982	2025-11-12
C0006	IN-2025-02960	2025-11-08

785 rows × 3 columns (first 6 shown)

Nth highest per group

"Second-highest paid employee in each department" (a top interview question):

SQL

```sql
WITH r AS (
    SELECT department, emp_id, monthly_income,
           DENSE_RANK() OVER (PARTITION BY department ORDER BY monthly_income DESC) AS rnk
    FROM employees
)
SELECT department, emp_id, monthly_income
FROM r
WHERE rnk = 2
ORDER BY department;
```

RESULT

department	emp_id	monthly_income
Finance	E00145	185500
HR	E00450	175750
IT	E01473	194500
Operations	E00378	184167
R&D	E00580	188083
Sales	E01218	184833

6 rows × 3 columns

Use `DENSE_RANK` for "Nth highest **value**" (ties share a position, no gaps) and `ROW_NUMBER` for "Nth **row**".

Removing duplicates with ROW_NUMBER

SQL

```sql
WITH d AS (
    SELECT *, ROW_NUMBER() OVER (PARTITION BY customer_id, order_date, product_id ORDER BY
item_id) AS rn
    FROM order_items_raw
)
DELETE FROM d WHERE rn > 1;            -- SQL Server lets you DELETE through a CTE
```

Rows with the same business key get numbers 1, 2, 3…; keeping `rn = 1` keeps one copy. Always check with a `SELECT … WHERE rn > 1` first.

Let's count how many lines in our data repeat the same product in the same order (what a de-duplication would touch):

SQL

```sql
WITH d AS (
    SELECT order_id, product_id,
           ROW_NUMBER() OVER (PARTITION BY order_id, product_id ORDER BY row_id) AS rn
    FROM superstore
)
SELECT COUNT(*) AS repeated_lines FROM d WHERE rn > 1;
```

RESULT

repeated_lines
70

1 rows × 1 columns

(These are legitimate repeat lines, not errors; the point is the pattern.)

NTILE: buckets and percentiles

SQL

```sql
WITH cust AS (
    SELECT customer_id, SUM(sales) AS sales FROM superstore GROUP BY customer_id
), q AS (
    SELECT customer_id, sales, NTILE(4) OVER (ORDER BY sales DESC) AS quartile FROM cust
)
SELECT quartile,
       COUNT(*)               AS customers,
       ROUND(SUM(sales), 0)   AS sales,
       ROUND(100.0 * SUM(sales) / SUM(SUM(sales)) OVER (), 1) AS pct_of_total_sales
FROM q
GROUP BY quartile
ORDER BY quartile;
```

RESULT

quartile	customers	sales	pct_of_total_sales
1	197	111,100,409	55.2
2	196	54,386,341	27.0
3	196	28,080,117	13.9
4	196	7,846,704	3.9

4 rows × 4 columns

Quartile 1 (the top 25% of customers) generates a disproportionate share of sales: the **Pareto principle** from the Excel charts lesson, now computed in SQL. NTILE is also how RFM scoring is done (ML track).

Aggregate window functions

Any aggregate (`SUM`, `AVG`, `COUNT`, `MIN`, `MAX`) becomes a window function with `OVER`. Without `ORDER BY` in the window, it uses the whole partition: perfect for totals, shares and comparisons with the

group average.

```sql
SELECT department, emp_id, monthly_income,
       ROUND(AVG(monthly_income) OVER (PARTITION BY department), 0)          AS dept_avg,
       monthly_income - ROUND(AVG(monthly_income) OVER (PARTITION BY department), 0) AS
vs_dept_avg,
       COUNT(*) OVER (PARTITION BY department)                               AS
dept_headcount,
       MAX(monthly_income) OVER (PARTITION BY department)                    AS dept_max
FROM employees
ORDER BY department, monthly_income DESC;
```

RESULT

department	emp_id	monthly_income	dept_avg	vs_dept_avg	dept_headcount	dept_
Finance	E00519	209833	83,681	126,152	154	2098
Finance	E00145	185500	83,681	101,819	154	2098
Finance	E00336	174417	83,681	90,736	154	2098
Finance	E00201	169500	83,681	85,819	154	2098
Finance	E01172	168167	83,681	84,486	154	2098
Finance	E01151	155167	83,681	71,486	154	2098
Finance	E00759	152083	83,681	68,402	154	2098
Finance	E01004	144667	83,681	60,986	154	2098

1,500 rows × 7 columns (first 8 shown)

This replaces the correlated subquery from Lesson 10 ("earns more than the department average") with one pass over the data.

GROUP BY vs window functions

	GROUP BY	Window function
Rows returned	One per group	Same as input
Detail columns	Only grouped columns	All columns
Use for	Summary reports	Rankings, running totals, shares, comparisons with group values
Filter the result	`HAVING`	Wrap in CTE/subquery and `WHERE`
Can combine	Yes: group first, then window over the groups	

Summary

KEY TAKEAWAYS

- `OFFSET n ROWS FETCH NEXT m ROWS ONLY` (with `ORDER BY`) paginates; it can skip rows, unlike `TOP`.
- A **window function** adds a calculation over a set of related rows to each row, without collapsing them. `OVER (PARTITION BY … ORDER BY …)` defines the window.
- `ROW_NUMBER` (unique), `RANK` (ties, gaps), `DENSE_RANK` (ties, no gaps), `NTILE(n)` (equal buckets).
- To filter on a window result, compute it in a CTE or derived table, then use `WHERE`.
- Top-N per group and Nth highest per group: `DENSE_RANK`/`ROW_NUMBER` partitioned by group; de-duplicate with `ROW_NUMBER … WHERE rn > 1`.
- Aggregate windows (`SUM() OVER (PARTITION BY …)`) show totals, shares and group averages beside each row.

Quick quiz

6 questions · tap an answer to check it

1. Scores 100, 90, 90, 80. What does DENSE_RANK() OVER (ORDER BY score DESC) return?

A 1, 2, 3, 4

B 1, 2, 2, 4

C 1, 2, 2, 3

D 1, 1, 2, 3

Answer: C. DENSE_RANK gives tied rows the same rank and continues without gaps.

2. Why can't you write `WHERE ROW_NUMBER() OVER (...) = 1`?

A ROW_NUMBER needs GROUP BY

B Window functions are evaluated after WHERE, so you must filter in an outer query or CTE

C OVER cannot contain ORDER BY

D It is allowed in SQL Server

Answer: B. Logical processing order: WHERE runs before SELECT, where windows are computed.

3. What is the difference between PARTITION BY and GROUP BY?

A None

B PARTITION BY defines the window groups but keeps every row; GROUP BY collapses rows

C PARTITION BY only works with RANK

D GROUP BY keeps all rows

Answer: B. Partitions restart the calculation for each group without reducing the number of rows.

4. Which returns the 3rd highest distinct salary overall?

- **A** DENSE_RANK() OVER (ORDER BY salary DESC) = 3
- **B** ROW_NUMBER() OVER (ORDER BY salary DESC) = 3
- **C** TOP 3 salary
- **D** RANK() OVER (ORDER BY salary) = 3

Answer: A. DENSE_RANK counts distinct values; ROW_NUMBER would count rows even if values repeat.

5. `SUM(sales) OVER (PARTITION BY region)` returns:

- **A** One row per region
- **B** On every row, the total sales of that row's region
- **C** The overall total on every row
- **D** A running total

Answer: B. Without ORDER BY in the window, it sums the whole partition.

6. To fetch rows 21–30 of a sorted result you write:

- **A** `TOP 30`
- **B** `ORDER BY … OFFSET 20 ROWS FETCH NEXT 10 ROWS ONLY`
- **C** `LIMIT 10, 20` in T-SQL
- **D** `WHERE ROW 21 TO 30`

Answer: B. Skip 20, take 10.

EXERCISE

Window practice

(1) For each customer segment, rank customers by total sales and show the top 3. (2) For each order line, show its sales as a percentage of its order's total. (3) For each department, list employees whose income is within the top 5% of their department (use NTILE(20)). (4) Show products whose total sales rank in the top 10% of their category.

Show solution (worked answer)

SQL

```sql
-- (1)
WITH t AS (SELECT c.segment, c.customer_name, SUM(oi.sales) AS sales
           FROM order_items oi JOIN orders o ON o.order_id = oi.order_id JOIN
customers c ON c.customer_id = o.customer_id
           GROUP BY c.segment, c.customer_id, c.customer_name),
     r AS (SELECT *, DENSE_RANK() OVER (PARTITION BY segment ORDER BY sales DESC) AS
rnk FROM t)
SELECT * FROM r WHERE rnk ⩽ 3 ORDER BY segment, rnk;
-- (2)
SELECT item_id, order_id, sales, ROUND(100.0 * sales / SUM(sales) OVER (PARTITION BY
order_id), 1) AS pct_of_order FROM order_items;
-- (3)
WITH x AS (SELECT department, emp_id, monthly_income, NTILE(20) OVER (PARTITION BY
department ORDER BY monthly_income DESC) AS bucket FROM employees)
SELECT * FROM x WHERE bucket = 1;
-- (4)
WITH p AS (SELECT pr.category, pr.product_id, SUM(oi.sales) AS sales FROM order_items
oi JOIN products pr ON pr.product_id = oi.product_id GROUP BY pr.category,
pr.product_id),
     n AS (SELECT *, NTILE(10) OVER (PARTITION BY category ORDER BY sales DESC) AS
decile FROM p)
SELECT * FROM n WHERE decile = 1;
```

INTERVIEW CORNER

Questions you may be asked

Q: RANK vs DENSE_RANK vs ROW_NUMBER? ROW_NUMBER numbers rows uniquely; RANK gives ties the same rank and skips following ranks; DENSE_RANK gives ties the same rank without gaps.

Q: Find the second-highest salary in each department. CTE with `DENSE_RANK() OVER (PARTITION BY dept ORDER BY salary DESC)`, filter `= 2`.

Q: How do you remove duplicate rows keeping one? `ROW_NUMBER() OVER (PARTITION BY business_key ORDER BY …)` and delete rows where it is greater than 1.

Q: Window function vs GROUP BY? Windows keep detail rows and add group-level values; GROUP BY returns one row per group.

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